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Introduction
to Ensemble
Learning

The Statistical
Theory

Why Ensemble
Learning?

The Structure
of an
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System

BAGging

Random
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Boosting

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Ensemble Learning

Srijit Mukherjee

Indian Statistical Institute

July 28, 2020

Nomenclature

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Dictionary Meaning

a group of items viewed as a whole rather than individually.

Ensemble Decision Making

Decision made based on(by) groups of items/people rather than individual item/people.

Democracy - Ensemble Decision Making

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Idea

A group of people vote to make a decision, whether to choose an elected official or to decide on a new law, is in fact based on ensemble-based decision making.



Figure: Vote

Ensemble Learning

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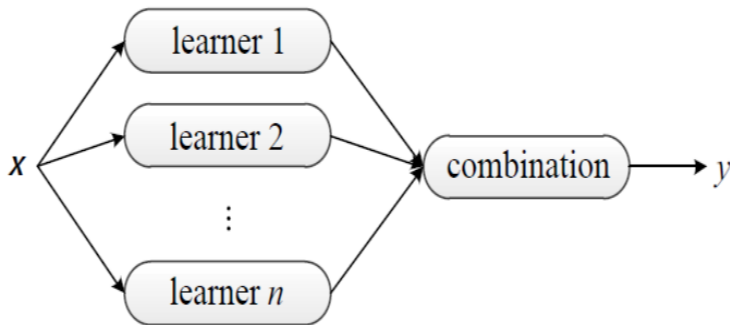


Figure: A Common Structure of Ensemble Learning

Example of Ensemble Learning

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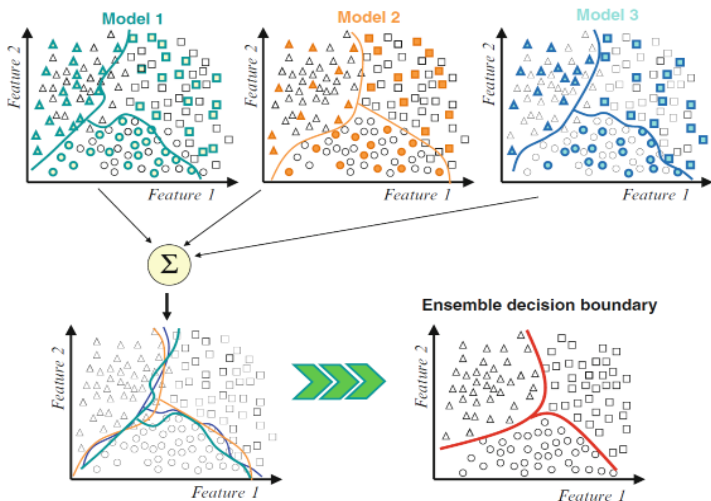
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A Basic Problem

The Setup

You have $X_1, \dots, X_n$ samples from $N(\theta, 1)$ distribution, where the mean of the distribution θ is unknown.

Learner_{*i*}

$$E(X_i) = \theta$$

$$MSE(X_i) = Var(X_i) = 1$$

Combination

$$E(\bar{X}) = \theta$$

$$MSE(\bar{X}) = Var(\bar{X}) = \frac{1}{n}$$

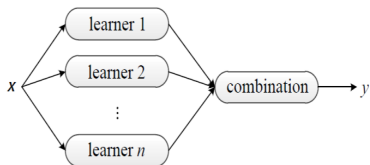


Figure: An Ensemble Structure

Algorithm = Statistic

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An Algorithm f acting on Data is a Random Variable : $f(Data)$

The Ensemble Idea

The Setup

You have $L_1, \dots, L_n$ algorithms, where we want to predict f .

Learner_{*i*}

$$\text{Bias}(L_i) = \epsilon \text{ and } \text{Var}(L_i) = \sigma^2$$

$$\text{MSE}(L_i) = \epsilon^2 + \sigma^2$$

Combination

$$\text{Bias}(\bar{L}) = \epsilon$$

$$\text{MSE}(\bar{L}) = \epsilon^2 + \text{Var}(\bar{L}) = \epsilon^2 + \frac{\sigma^2}{n}$$

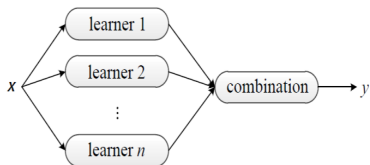


Figure: An Ensemble Structure

The Correlated Ensemble Idea

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The Setup

You have $L_1, \dots, L_n$ algorithms, where we want to predict f .
 $cor(L_i, L_j) = \rho$.

Combination

$$Bias(\bar{L}) = \epsilon$$

$$MSE(\bar{L}) = \epsilon^2 + Var(\bar{L})$$

$$Var(\bar{L}) = \sigma^2 \times \frac{\rho(n-1) + 1}{n}$$

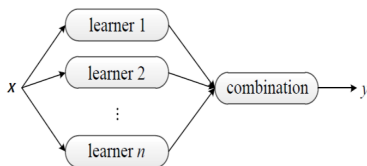


Figure: An Ensemble Structure

Moral of the Theory

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$\operatorname{argmin}_{\rho} \operatorname{MSE}(\bar{L}) = \operatorname{argmin}_{\rho} \operatorname{Var}(\bar{L}) = 0 \implies L'_i\text{'s are preferred to be uncorrelated (negatively correlated*) for optimum performance.}$

Bootstrapping

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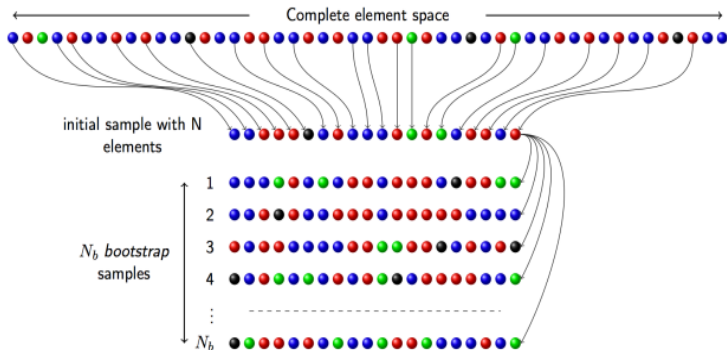
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The Decision Tree Fatigue

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The Instability Theorem

Assume that t_0 is the split of a node and t is a split that is almost as good as t_0 with respect to a fraction c . Then t may be the split selected to partition the changed data, if a change is made to a fraction $\frac{c \cdot |D|}{|S|}$ of the objects in the training sample S , where D is the data associated with the node.

nyears	sports car	risk
1	yes	high
2	no	high
2	yes	high
4	no	low
8	no	low



original sample and the corresponding tree

nyears	sports car	risk
1	yes	high
4	yes	high
2	yes	high
4	no	low
8	no	low



changed sample and the corresponding tree

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A Hypothetical Example

If you take that one example, it could tip the splitting criterion in a different direction. For example, imagine that you have two attributes you can split on: attribute A (gain of 0.71) and attribute B (gain of 0.7). Imagine you're at the root node. Clearly in this case we would split on A. Now let's take one example out. Now the gain of Attribute A is 0.69. Now you split on B. And the entire tree now is completely different because of one example.

Example

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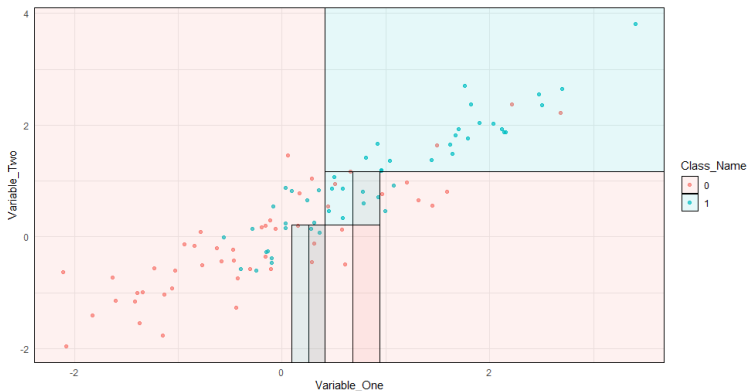
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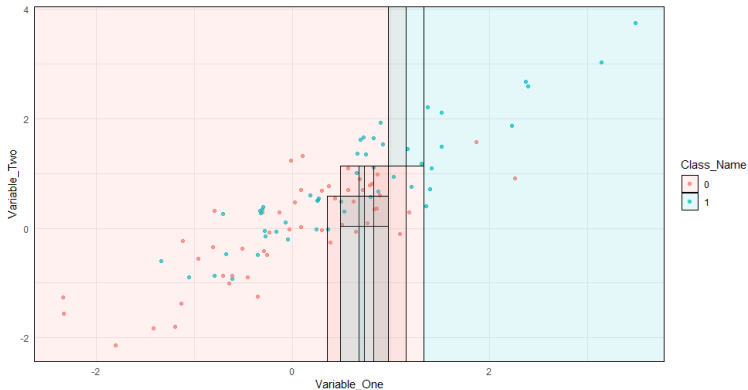
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The Messi MSE

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Mean Square Error

$$E \left(y_0 - \hat{f}(x_0) \right)^2 = \text{Var} \left(\hat{f}(x_0) \right) + \left[\text{Bias} \left(\hat{f}(x_0) \right) \right]^2 + \text{Var}(\epsilon)$$

Bias and Variance

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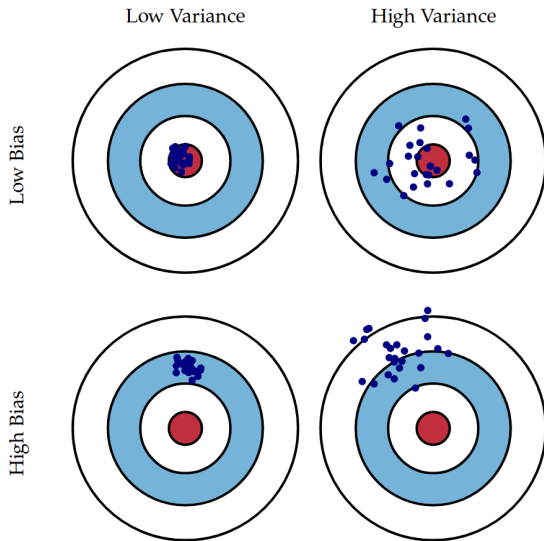
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Bias Variance Trade off

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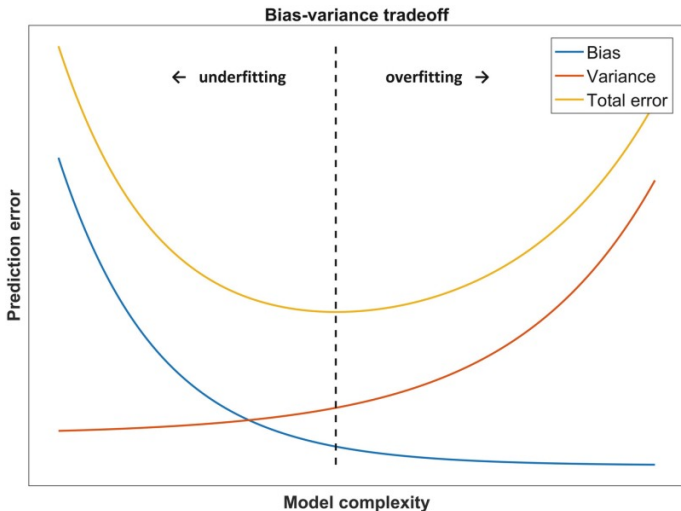
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An Effective Ensemble System

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The Three Pillars

- Data Sampling/Selection
- Training Member Learners L_i
- Combining Learners L_i to give L

Pillar One: Data Sampling/ Selection

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Keywords

- **Diversity** (We need diversity in the decisions of ensemble members.)
- **Independent, Negative Correlation** (Ideally, learner's outputs should be independent or preferably negatively correlated.)

Strategies

- Different subsets of the training data
- Bootstrapped replicas of the training data (Bagging).
- Sampling from a distribution that favors previously misclassified samples (Boosting).
- Use different subsets of the available features to train each learner (Random Sub Space Method)

Pillar Two: Training Member Learners L_i

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Strategies

- Bagging (and related algorithms arc-x4 and random forests)
- Boosting (and its many variations)
- Stack Generalization
- Hierarchical MoE

Pillar Three: Combining Learners L_i to give L

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Discrete Output Combination Strategies

- Majority Voting
- Weighted Majority Voting
- Borda Count (by Jean Charles de Borda, 1770)

Continuous Output Combination Strategies

- Mean Rule
- Weighted Average
- Trimmed mean
- Product Rule
- Generalized Mean
- Decision Template

Bootstrap AGGregation Algorithm (by Breiman)

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Keywords

- Bootstrap
- Aggregation
- Parallel

Diversity

The diversity in the ensemble is ensured by the variations within the bootstrapped replicas on which each learner is trained, as well as by using a relatively **weak learner** whose **decision boundaries measurably vary with respect to relatively small perturbations in the training data.**

The Diagram

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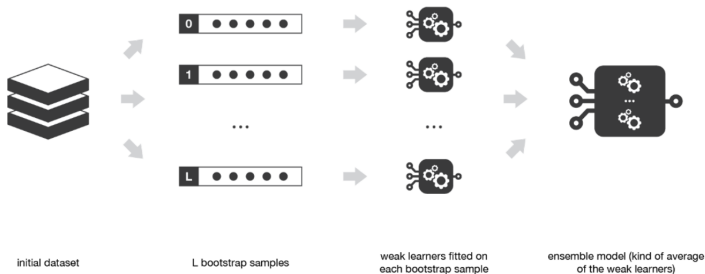
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The Demand of the Algorithm

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Inputs to Algorithm

- A training data set S of cardinality N .
- A Supervised learning Algorithm
- Base Weak Learner
- Integer T specifying ensemble size.
- Percent R to create bootstrapped training data.

The Algorithm (Classifier)

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Do $t = 1, \dots, T$

Pillar One: Data Sampling / Selection

Take a bootstrapped replica S_t by randomly drawing $R\%$ of S .

Pillar Two: Training Learning Members

- Call Base Learner with S_t and receive the hypothesis h_t .
- Add h_t to the ensemble, $\mathcal{E} \leftarrow \mathcal{E} \cup h_t$.

The Algorithm (Classifier)

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Pillar Three: Combining Learners

- Evaluate the ensemble $\mathcal{E} = \{h_1, \dots, h_T\}$ on x
- Let $v_{t,c} = 1$ if h_t chooses class ω_c , and 0, otherwise.
- Obtain total vote received by each class

$$V_c = \sum_{t=1}^T v_{t,c}, c = 1, \dots, C$$

Output

Class with the highest V_c

When to BAgg? Why to BAgg?

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Why to BAgg?

Bagging decreases variance of the base model without changing the bias, thus more stable.

When and When not to BAgg?

- High Bias (Avoid)
- Over fitting $\implies$ High Variance (Use)
- Low Bias, High Variance (Use)

The Algorithm

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Keywords

- Bagging
- Decision Tree
- Decorrelate

The Diagram

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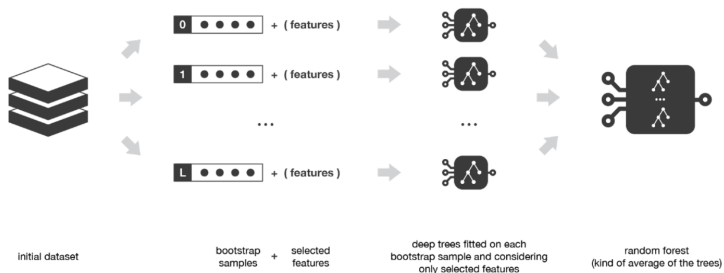
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The Algorithm

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Input

Let $D = \{(x_1, y_1), \dots, (x_N, y_N)\}$ denote the training data, with $x_i = (x_{i,1}, \dots, x_{i,p})^T$.

For $j = 1$ to J :

The Algorithm

- Take a bootstrap sample D_j of size N from D .
- Using the bootstrap sample D_j as the training data, fit a tree using Decision Tree.
 - Start with all observations in a single node.
 - Repeat the following steps recursively for each unsplit node until the stopping criterion is met

The Algorithm

The Steps

- 1 Select $m = \sqrt{p}$ or $\log_2 p$ predictors at random from the p available predictors.
- 2 Find the best binary split among all binary splits on the m predictors from Step 1.
- 3 Split the node into two descendant nodes using the split from Step 2.

The Prediction

To make a prediction at a new point x , $\hat{f}(x) = \frac{1}{J} \sum_{j=1}^J \hat{h}_j(x)$ for regression $f(x) = \arg \max_y \sum_{j=1}^J I(\hat{h}_j(x) = y)$ for classification where $\hat{h}_j(x)$ is the prediction of the response variable at x using the j th tree

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Why Random Forest?

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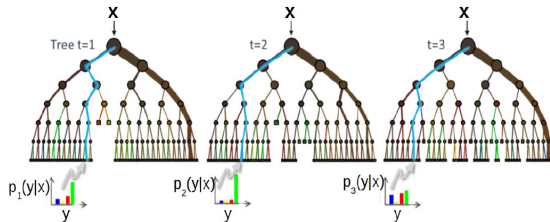
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Why?

- I can stare at it all day long.
- Missing Value Imputation
- De Correlate
- Variable Importance
- Regression and Classification

Boost is The Secret of My (Our) Energy

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Boosting: Adaboost (by Freund and Schapire)

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Keywords

- Series
- Adaboost
- Bootstrap (on iteratively updated training data set)
- Weighted Voting (Aggregation / Combination)

The Diagram

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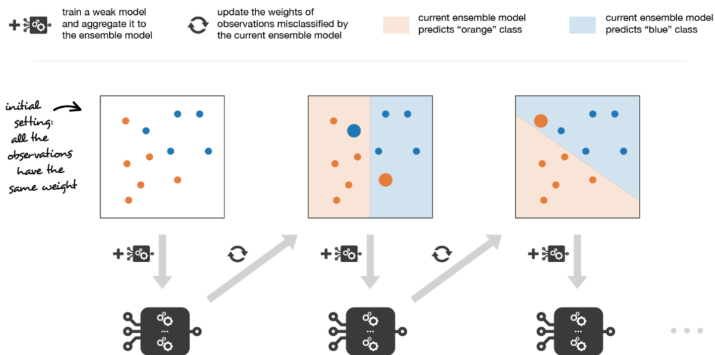
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Inputs to Algorithm

- Training data = $\{x_i, y_i\}, i = 1, \dots, N, y_i \in \{\omega_1, \dots, \omega_C\}$,
- Supervised learner
- Base Learner
- Ensemble size T
- Initialize $D_1(i) = 1/N$

Do $t = 1, \dots, T$

Pillar One: Data Sampling / Selection

Draw training subset S_t from the distribution D_t

The Algorithm (Classifier)

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Do $t = 1, \dots, T$

Pillar Two: Training Learning Members

- Train Base Learner on S_t , receive hypothesis $h_t : X \rightarrow Y$
- Calculate the error of h_t :

$$\varepsilon_t = \sum_i I[h_t(x_i) \neq y_i] D_t(x_i)$$

If $\varepsilon_t > 1/2$ abort.

- Set

$$\beta_t = \varepsilon_t / (1 - \varepsilon_t)$$

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Pillar Two: Training Learning Members

- Update sampling distribution

$$D_{t+1}(i) = \frac{D_t(i)}{Z_t} \cdot \begin{cases} \beta_t, & \text{if } h_t(x_i) = y_i \\ 1, & \text{otherwise} \end{cases}$$

where $Z_t = \sum_i D_t(i)$ is a normalization constant to ensure that D_{t+1} is a proper distribution function.

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Pillar Three: Combining Learners with Output

Weighted Majority Voting: Given unlabeled instance z , obtain total vote received by each class

$$V_c = \sum_{t: h_t(z) = \omega_c} \log \left(\frac{1}{\beta_t} \right), \quad c = 1, \dots, C$$

Output: Class with the highest V_c

Note

The reciprocals of the normalized errors of individual classifiers are used as voting weights in weighted majority voting in AdaBoost.M1; hence, classifiers that have shown good performance during training (low β_t) are rewarded with higher voting weights.

Boosting for Regression Trees

- 1 Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all i in the training set.
- 2 For $b = 1, 2, \dots, B$, repeat:
 - (a) Fit a tree $\hat{f}^b$ with d splits ($d + 1$ terminal nodes) to the training data (X, r)
 - (b) Update $\hat{f}$ by adding in a shrunk version of the new tree:

$$\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x)$$

- (c) Update the residuals,

$$r_i \leftarrow r_i - \lambda \hat{f}^b(x_i)$$

- 3 Output the boosted model,

$$\hat{f}(x) = \sum_{b=1}^B \lambda \hat{f}^b(x)$$

Why to Boost?

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Why to Boost?

The base learners in boosting are weak learners in which the bias is high, and the predictive power is just a tad better than random guessing. Each of these weak learners contributes some vital information for prediction, enabling the boosting technique to produce a strong learner by effectively combining these weak learners. The final strong learner brings down both the bias and the variance.

When to Boost?

High Bias, Low Variance (preferred)

The Communist Adaboost

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● random forest ● adaboost ● ensemble methods



Don't blame that Poor Guy

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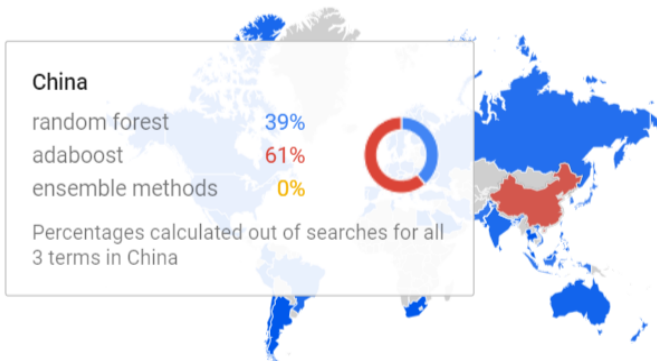
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H_0 : Random Forest is important.

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We don't have enough data to reject H_0 .

